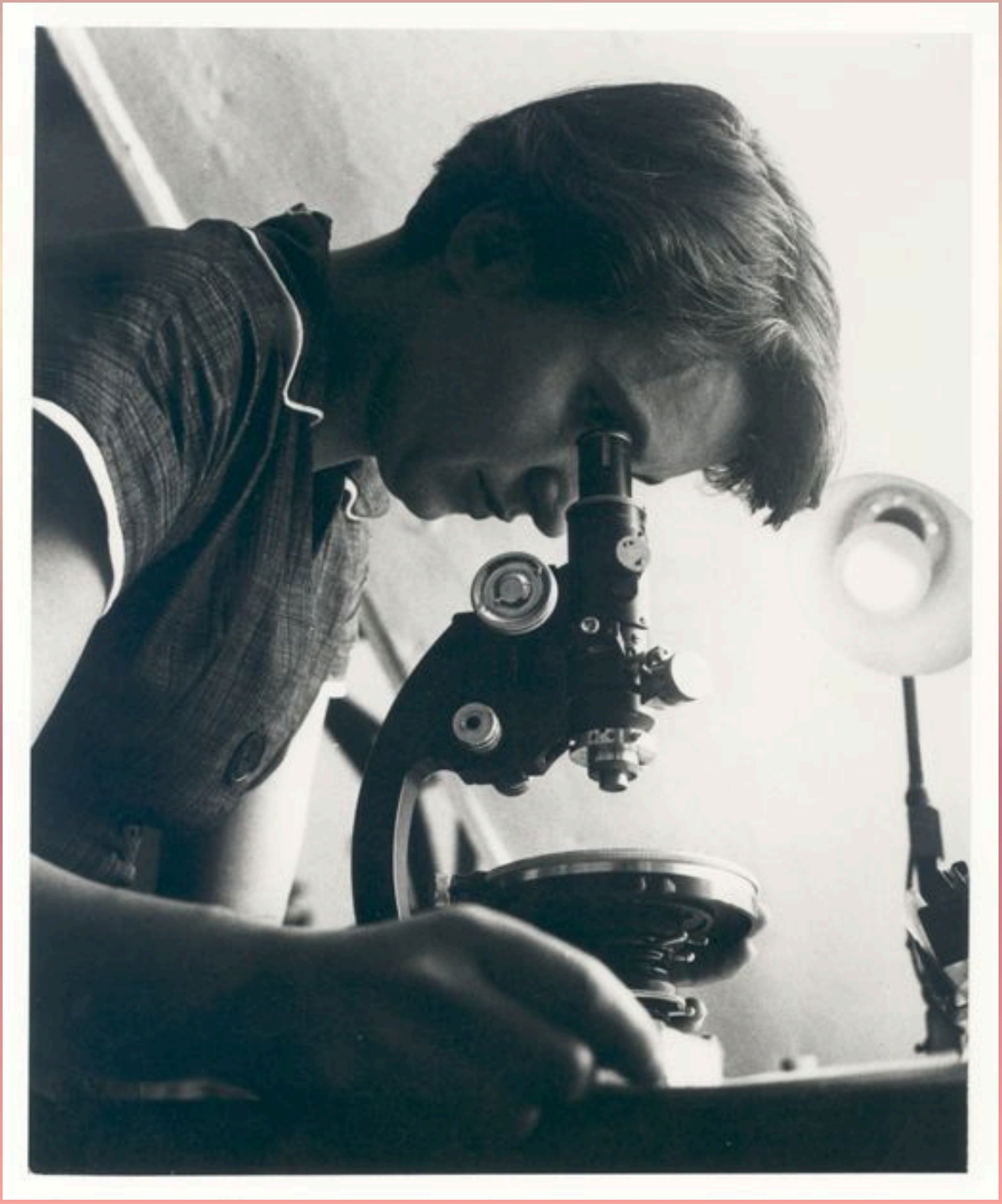
INAUGURAL ISSUE NOVEMBER 2026

FabScientist



THE WOMAN WHO CHANGED HOW WE SEE LIFE

RROSALIND FRANKLIN AND THE STORY BEHIND THE DISCOVERY OF DNA'S STRUCTURET

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SCIENCE . PEOPLE . DISCOVERY

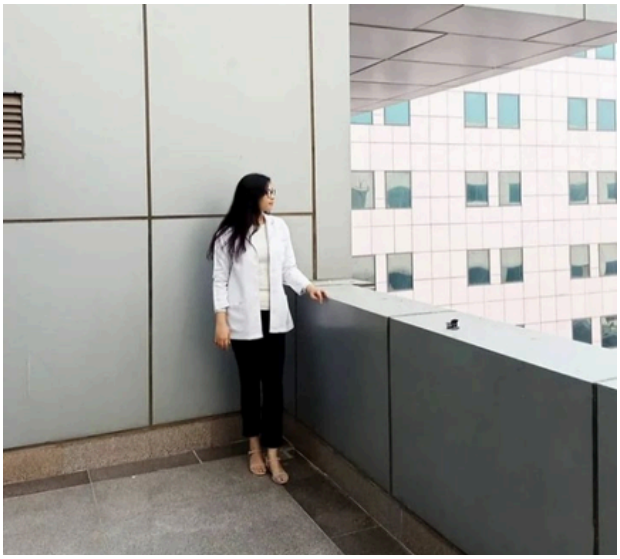
EDITOR'S LETTER

NOVEMBER/2026

Science has always been more than a collection of facts to me. Finding fascinating stories behind scientific discoveries and the mind behind them has always inspired me. It is a way of looking at the world, of asking questions, challenging what we think we know. Yet, somewhere between the laboratory and the pages of scientific journals, the wonder behind discovery can sometimes get lost. The people, the perseverance, the unexpected turns and the questions that drive scientific progress deserve to be told, too.

This magazine is a small beginning, built around a simple belief that meaningful science communication can bring research closer to everyday life. Through stories of remarkable scientists, discoveries that have changed our understanding of the world, and ideas that continue to shape our future, I hope to create a space where curiosity feels welcome, regardless of one's scientific background.

"FabScientist was born from this thought: that science should not only inform us, but inspire us to look closer."



Our inaugural cover story features Rosalind Franklin, whose work on X-ray diffraction contributed crucial evidence to the discovery of DNA's structure. Her story invites us to look beyond a scientific breakthrough and consider the people, the work and the history behind it.

As FabScientist takes its first steps, I hope these pages encourage you to ask questions, explore unfamiliar ideas and see science with renewed curiosity.

Every meaningful journey begins somewhere. This is ours.

Welcome to FabScientist.

Kritika Sonar

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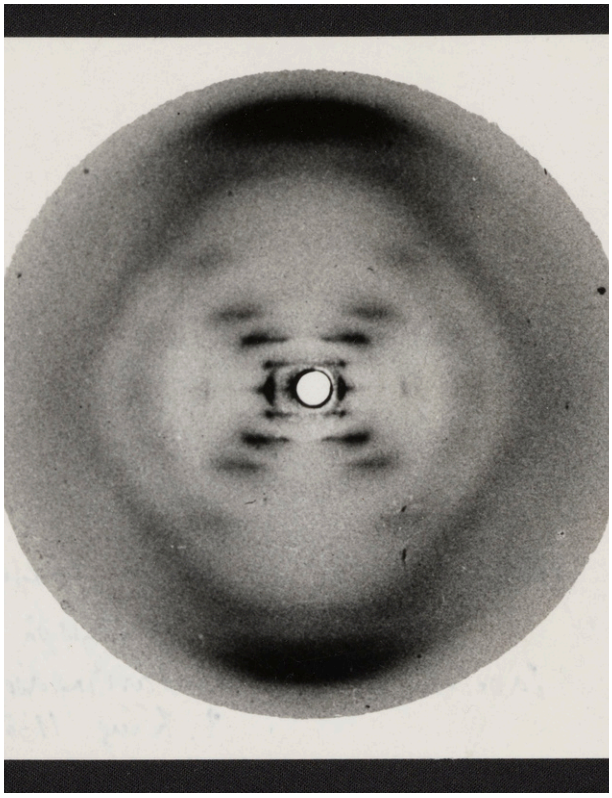
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THE X IN THE PHOTOGRAPH



THE X IN THE PHOTOGRAPH

Rosalind Franklin started her career as a chemist before becoming a DNA researcher studying Natural sciences at Cambridge University. At a time when women were facing barriers in scientific careers. She spent years investigating coal and graphite. But Paris changed the course of her life. There, under the guidance of crystallographer Jacques Mering, she developed her expertise in X-ray diffraction. This later on became central to her study of DNA structure.

In the early 1950s, Rosalind Franklin was accepted into King's College London, where she redirected her focus towards learning and studying DNA. She could not see the molecule itself. No microscope could simply show her its shape. DNA was too small, too intricate, too far beyond the reach of ordinary vision. So, she had to find another way. She turned to X-rays.

THE MAKING OF PHOTOGRAPH 51

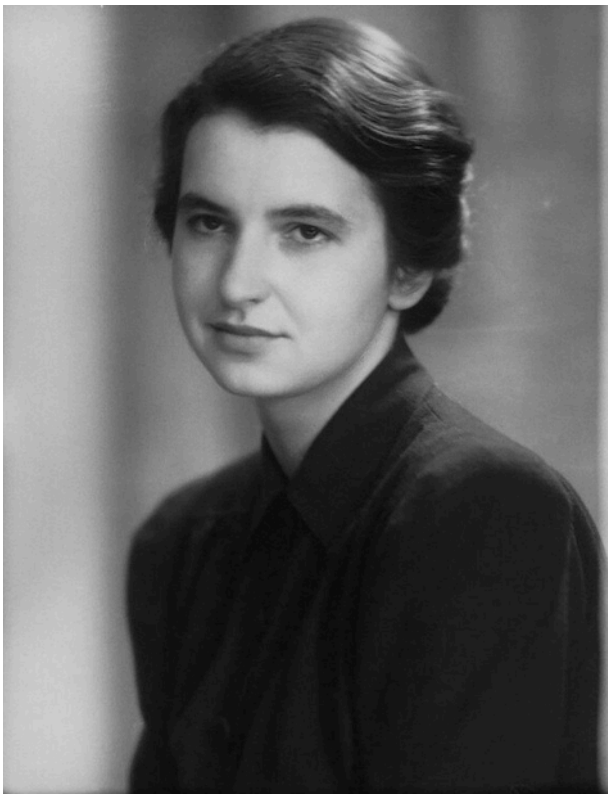
When X-rays pass through a carefully prepared sample of DNA, they scatter. Those scattered rays leave behind a pattern on photographic film, and Franklin had spent years learning how to read such patterns. The image, which was number 51 in a series taken with varying humidity and water content, was the best of the high-humidity, "wet," or "B-form" photos. A distinct "A-form" pattern was observed with lower humidity, more akin to the diffraction photos obtained by Maurice Wilkins before she arrives at King's.

THE NUMBERS BEHIND THE MYSTERY

Among the clues were distances measured in ångströms a unit so small that ten million ångströms would fit into a single millimetre. Franklin's data pointed to a repeating distance of about 34 ångströms, with components spaced roughly 3.4 ångströms apart.
These numbers mattered. They suggested that the molecule had a regular, repeating architecture. The X suggested a helix. The measurements told scientists how that helix was arranged.

A RACE TO UNDERSTAND DNA

While Franklin was studying DNA experimentally, James Watson and Francis Crick were trying to determine its structure through model building.
They were working from experimental evidence, including X-ray diffraction results, to ask what kind of molecular structure could possibly fit all the observations. In 1953, their famous double-helical model was published in Nature. Franklin and Raymond Gosling published their own experimental findings in the same issue. Franklin's work was crucial. So were the contributions of Gosling, Wilkins, Watson, Crick and many others. She did not publish the double-helix model herself, and we should be careful not to turn her into a scientist who merely “had the answer stolen from her. “Her real achievement is more interesting. She was an extraordinarily careful experimental scientist who helped establish the physical evidence from which the structure of DNA could be understood.

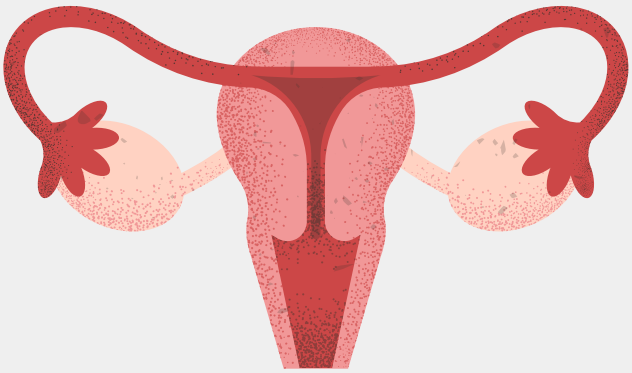


What happened to Rosalind Franklin? Franklin died in 1958, at just 37. By then, the double helix had already become one of the most recognisable structures in science. In the years that followed, her contribution became the subject of considerable debate. Popular retellings often reduced the story to a simple injustice: a brilliant woman produced a photograph, and two men took the discovery from her. The real history is more complicated.

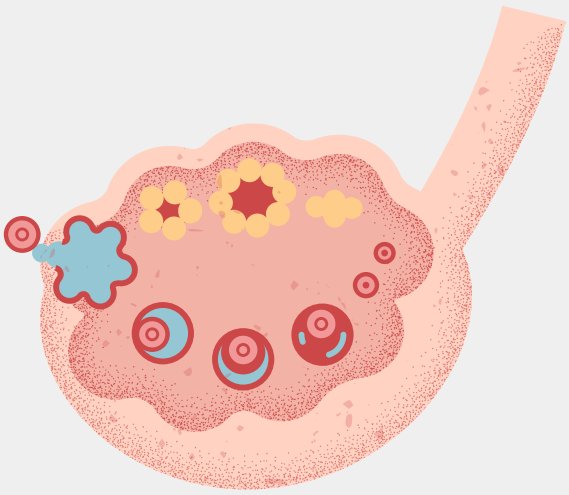
THE FORGOTTEN NETWORK

INSIDE THE OVARY

Rete ovarii, Latin for “ovarian network”, is one of those structures hidden in plain sight.
Located within the ovarian hilum, the rete ovarii is a small network of epithelial tubules that is considered the female counterpart of structures found in the developing testis. It is a remnant of the embryonic reproductive system, formed from tissue associated with the embryonic gonad.



For decades, structures such as the rete ovarii received little attention compared with the more familiar parts of the ovary: follicles, oocytes, the corpus luteum and the surrounding stroma. Yet the rete ovarii offers a glimpse into something fundamental—the fact that the reproductive system is built through a complex process of embryonic development in which tissues acquire, retain and sometimes lose their original identities.



The rete ovarii is generally considered a vestigial structure, meaning that no essential reproductive function has been firmly established for it in humans. But “vestigial” does not mean irrelevant. Its anatomy and development have helped researchers understand the organization of the ovary and the evolutionary relationship between male and female reproductive structures.

Even more intriguingly, remnants of the embryonic reproductive ducts and tissues can sometimes become clinically significant later in life, forming unusual cysts or contributing to rare pathological conditions.
The rete ovarii is therefore a reminder that anatomy is not simply a catalogue of organs that perform obvious functions. Hidden within the body are traces of developmental history—small structures that tell us where we came from, how the reproductive system was assembled, and how much of human biology remains to be understood.

CHANGING THE WAY WE LEARN AND UNDERSTAND .

The 2026 Nobel Prize in Physiology or Medicine

The history of the 2026 Nobel Prize in Physiology or Medicine in the new field of optogenetics goes back to the 1970s. In an article in Scientific American, Nobel Laureate Francis Crick famously stated that to study the brain signals, we had to come up with an alternative. The old tools targeted every cell in the brain, and the chemical drugs lacked precision. In 1999, he wrote that the ideal signal would be light but said it was a far-fetched idea at the time. This eventually led to the development of a new field of optogenetics.

How does optogenetics work?

The human brain is made up of billions of cells; therefore, the technique uses light to control brain cells, wherein flashes of light stimulate a particular nerve with millisecond precision.

Peter Hegemann and Georg Nagel, microbiologists at Humboldt University of Berlin, Germany and Nagel University of Würzburg, discovered the channelrhodopsin, a light-sensitive protein found in algae. They found that when channelrhodopsin is exposed to blue light, it acts as a physical gate or an ion channel that allows positive ions to flow and generate an electrical impulse.

Karl Deisseroth (The Neuroscientist) from Stanford University inserted the algal channelrhodopsin gene into the mammalian (rat) neurons. He used optical fibres to deliver blue light into living mice brains.When activated by blue light, channelrhodopsin opens ion channels and triggers electrical impulses.



Illuminating the brain

It helps in brain mapping and establishing relationships between neural activity and behaviour for understanding neurological disorders such as Parkinsons disease, schizophrenia, depression and epilepsy. It helps researchers go beyond to understand how neural networks work and react to every change in emotion. What once seemed like a far-fetched idea is now a new reality for understanding the organ that makes us who we are.



DID YOU KNOW?



DID YOU KNOW?
YOUR RED BLOOD CELLS HAVE NO NUCLEUS.
MATURE HUMAN RED BLOOD CELLS LOSE THEIR NUCLEI AS THEY DEVELOP, LEAVING MORE ROOM FOR HAEMOGLOBIN, THE PROTEIN THAT CARRIES OXYGEN AROUND YOUR BODY.



DID YOU KNOW?
A DAY ON VENUS IS LONGER THAN ITS YEAR.
VENUS TAKES ABOUT 243 EARTH DAYS TO ROTATE ONCE ON ITS AXIS, BUT ONLY ABOUT 225 EARTH DAYS TO ORBIT THE SUN.



DID YOU KNOW?
TINY ANIMALS CAN SURVIVE EXTREME CONDITIONS.
TARDIGRADES, ALSO CALLED WATER BEARS, CAN ENTER A DORMANT STATE THAT HELPS THEM WITHSTAND EXTREME DEHYDRATION, VERY LOW TEMPERATURES AND EVEN THE VACUUM OF SPACE UNDER CERTAIN CONDITIONS.



DID YOU KNOW?
LOOKING INTO SPACE MEANS LOOKING INTO THE PAST.
LIGHT TAKES TIME TO TRAVEL. WHEN TELESCOPES OBSERVE DISTANT GALAXIES, THEY SEE LIGHT THAT BEGAN ITS JOURNEY MILLIONS OR EVEN BILLIONS OF YEARS AGO.



DID YOU KNOW?
A 150-MILLION-YEAR-OLD BIRD FOSSIL FROM CHINA, YUEZHUORNIS YANGYUANENSIS, HAD REMARKABLY SHORT WINGS, HINTING THAT EARLY BIRDS MAY HAVE EVOLVED MORE DIVERSE BODY FORMS THAN SCIENTISTS ONCE THOUGHT.



DID YOU KNOW?
THE WHITE DWARF WD J1330+6435 APPEARS CHARTREUSE GREEN, OFFERING CLUES ABOUT HOW SODIUM AND OTHER ELEMENTS IN A STAR'S ATMOSPHERE CAN INFLUENCE ITS COLOUR.

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